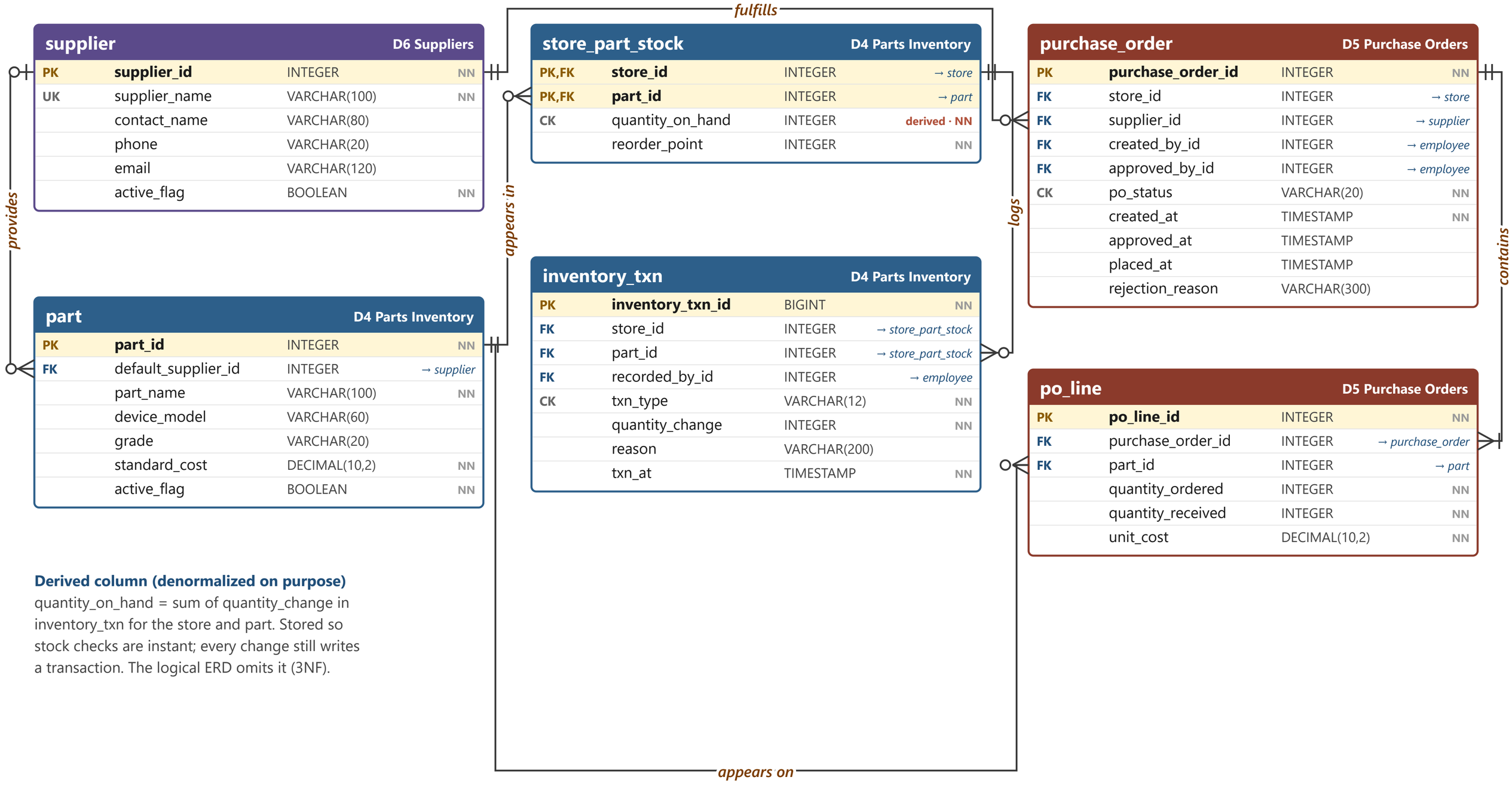
QuickPhone — Physical ERD detail 3 of 3: Parts, purchasing and suppliers

D4 Parts Inventory · D5 Purchase Orders · D6 Suppliers. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null



supplier — D6 Suppliers

PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

store_part_stock — D4 Parts Inventory

PK,FK	**store_id**	INTEGER	→ *store*
PK,FK	**part_id**	INTEGER	→ *part*
CK	quantity_on_hand	INTEGER	derived · NN
	reorder_point	INTEGER	NN

purchase_order — D5 Purchase Orders

PK	**purchase_order_id**	INTEGER	NN
FK	store_id	INTEGER	→ *store*
FK	supplier_id	INTEGER	→ *supplier*
FK	created_by_id	INTEGER	→ *employee*
FK	approved_by_id	INTEGER	→ *employee*
CK	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	

part — D4 Parts Inventory

PK	**part_id**	INTEGER	NN
FK	default_supplier_id	INTEGER	→ *supplier*
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

inventory_txn — D4 Parts Inventory

PK	**inventory_txn_id**	BIGINT	NN
FK	store_id	INTEGER	→ *store_part_stock*
FK	part_id	INTEGER	→ *store_part_stock*
FK	recorded_by_id	INTEGER	→ *employee*
CK	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

po_line — D5 Purchase Orders

PK	**po_line_id**	INTEGER	NN
FK	purchase_order_id	INTEGER	→ *purchase_order*
FK	part_id	INTEGER	→ *part*
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

Derived column (denormalized on purpose)

quantity_on_hand = sum of quantity_change in inventory_txn for the store and part. Stored so stock checks are instant; every change still writes a transaction. The logical ERD omits it (3NF).